

TIERRA DOUGHTY

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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on programming experience across robotics platforms, including Boston Dynamics Spot, and Aldebaran Robotics Pepper and NAO, plus a year translating complex technical systems into simpler concepts as a full-time STEAM educator and Lego robotics coach. Built and deployed AI-powered platforms using Flask, Google Vertex AI, and the OpenAI API, with a modular architecture designed for scalability. Drawn to the intersection of autonomous systems, computer vision, and distributed infrastructure, and excited to bring that focus to Amazon Robotics.

ROBOTICS & LEADERSHIP EXPERIENCE

LEGO Robotics Club Volunteer Coach

October 2025 - Present

Sacred Heart School, Lake Worth, FL

- Relaunched a dormant FIRST LEGO League robotics team after years of inactivity, recruiting and coaching 18 students in grades 4-8 with 100% retention across the season, plus a waitlist of 10+ additional students capped by room and equipment availability
- Designed and delivered a full curriculum using LEGO Mindstorms EV3, teaching block-based programming, sensor calibration, and motor control to students with no prior robotics experience
- Guided students through hands-on object detection projects using EV3 sensors, building foundational skills in autonomous behavior and robot-environment interaction during weekly meetings
- Organized a field trip to a FIRST LEGO League competition to expose students to competitive standards ahead of the team's first competitive season, once all 18 students had built foundational programming skills from scratch

Robotics Club Spokesperson and Programmer

September 2022 - May 2024

Saint Leo University, Saint Leo, FL

- Programmed a NAO robot in Python to autonomously identify a red ball via computer vision and execute a kicking motion into a goal, applying object detection and motion control fundamentals
- Developed conversational and expressive behaviors for Ferdinand, a robot head platform, using a Kotlin/Java SDK, programming approximately 50 lines of code to drive real-time facial expressions and voice interaction
- Explored autonomous locomotion control on Boston Dynamics Spot using C++, and demonstrated Spot, Pepper, and NAO robots for live public audiences, translating autonomous navigation and sensor-driven behavior into accessible presentations

AI AND TECHNICAL PROJECTS

Mindspark, AI-Powered Education Platform

January 2024 - Present

Computer Science Capstone Project

- Built and deployed a production-ready AI platform using Flask, Google Vertex AI, and the OpenAI API that generates personalized student learning plans in real time, with a modular architecture designed to swap AI models without rebuilding the system
- Iterated on prompt design across multiple model versions, measurably improving the relevance and accuracy of AI-generated content through systematic testing and refinement

Peer Pathways, AI-Enhanced Rideshare Platform

January 2026 - Present

Independent Project

- Spearheaded and developed a production-quality rideshare platform using React, Node.js, Express.js, MongoDB, and AI (for data scraping) that allows users to view networking events in their local area and carpool with professionals
- Designed and produced a community-focused carpooling web application to connect professionals who cannot afford rideshare costs to networking events and opportunities, currently iterating on wireframes and backend architecture using Claude AI Code workflow

TECHNICAL SKILLS

- Languages:** C++, Java, Python, JavaScript, SQL, HTML, CSS
- Robotics & Frameworks:** NAOqi, Java Swing, JavaFX, Tkinter, Flask, React, Django
- Cloud & Data:** Google Vertex AI, OpenAI API, MongoDB, SQL/NoSQL, exposure to AWS SDK/CLI
- AI-Assisted Development:** LLM integration, prompt design, Claude/ChatGPT for dev productivity workflows
- Tools:** Git/GitHub, RESTful APIs, Agile/CI-CD principles
- Operating Systems:** Linux, ROS (Robot Operating System), macOS, Windows

PROFESSIONAL EXPERIENCE

STEAM and Robotics Lead Teacher

August 2024 - Present

Sacred Heart School, Lake Worth, FL

- Advocated for and built a STEAM and robotics program from the ground up, including curriculum design, grant applications, lab setup, and supply procurement, then delivered it to 100+ students weekly across three grade levels daily and increasing student engagement by 80%
- Collaborated with fellow teachers to align STEAM content with existing curricula, while serving as the campus's assistant IT point of contact, reducing technology downtime by 40% through a structured troubleshooting system

Computer Science and Math Tutor

March 2022 - May 2024

Saint Leo University, Saint Leo, FL

- Taught data structures, algorithms, object-oriented principles, and system design fundamentals in Java and C++, while debugging student code across 50+ weekly sessions
- Increased student retention by 64% and improved coding assessment performance by 45% by analyzing engagement data monthly and redesigning instruction around individual learning gaps

EDUCATION

Bachelor of Science, Computer Science

Saint Leo University, Saint Leo, FL — Minor in Art, Cum Laude, GPA 3.81